

From topology to combinatorics via computability theory

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Summer School on "Reverse Mathematics: New Paradigms"
ESI · Vienna · July 2025

Recap

GST₁. Every infinite T_1 CSC space has an infinite subspace that is discrete or has the cofinite topology.

SRT₂²: Every stable coloring $c : [\mathbb{N}]^2 \rightarrow 2$ has an infinite homogeneous set.

D₂²: Every stable coloring $c : [\mathbb{N}]^2 \rightarrow 2$ has an infinite limit homogeneous set.

COH: Every sequence $\vec{R} = \langle R_n : n \in \mathbb{N} \rangle$ of sets has an infinite $\vec{R}$ -cohesive set.

Theorem (Cholak, Jockusch, and Slaman) Over RCA_0 , $\text{RT}_2^2 \leftrightarrow \text{SRT}_2^2 + \text{COH}$.

Lecture 3

Stability for other principles

Hirschfeldt & Shore: **general practice** of studying stable and cohesive parts.

In a poset $(\mathbb{N}, \leq_P)$, say x is

► **small** if $(\forall^\infty y)[x \leq_P y]$ ► **large** if $(\forall^\infty y)[x \geq_P y]$ ► **isolated** if $(\forall^\infty y)[x \not\leq_P y]$.

A linear order $(\mathbb{N}, \leq_L)$ is **stable** if every element is $\leq_L$ -small or $\leq_L$ -large.

(So $\leq_L$ has order type ω , ω^* , or $\omega + \omega^*$.)

A partial order $(\mathbb{N}, \leq_P)$ is **stable** if every element is $\leq_P$ -small or $\leq_P$ -isolated,
or every element is $\leq_P$ -large or $\leq_P$ -isolated.

$(\mathbb{N}, \leq_P)$ is **weakly stable** if every element is $\leq_P$ -small, $\leq_P$ -large, or $\leq_P$ -isolated.

Theorem (Jockusch, Kastermans, Lempp, Lerman, and Solomon).

Over RCA_0 , CAC for stable posets $\leftrightarrow$ CAC for weakly stable posets.

Cohesiveness for other principles

SRT_2^2 : Every stable coloring $c : [\mathbb{N}]^2 \rightarrow 2$ has an infinite homogeneous set.

SADS: Every stable linear order $\leq_L$ on $\mathbb{N}$ has an infinite ascending sequence or descending sequence.

SCAC: Every stable partial order $\leq_P$ on $\mathbb{N}$ has an infinite chain or antichain.

COH : Every sequence $\vec{R} = \langle R_n : n \in \mathbb{N} \rangle$ of sets has an infinite $\vec{R}$ -cohesive set.

CRT_2^2 : For every $c : [\mathbb{N}]^2 \rightarrow 2$ there is an infinite set X such that $c \upharpoonright [X]^2$ is stable.

Theorem (CJS). Over $\text{RCA}_0 + \text{B}\Sigma_2^0$, $\text{COH} \leftrightarrow \text{CRT}_2^2$.

We can define **CADS** and **CCAC** analogously to CRT_2^2 . This immediately yields CJS-style decomposition for ADS and CAC.

$$(\text{ADS} \leftrightarrow \text{SADS} + \text{CADS} \text{ and } \text{CAC} \leftrightarrow \text{SCAC} + \text{CCAC}).$$

Curiously, Hirschfeldt & Shore showed that $\text{CCAC} \leftrightarrow \text{ADS}$.

Stability for topological spaces

Consider a CSC space $\langle \mathbb{N}, \mathcal{U}, k \rangle$ with $\mathcal{U} = \langle U_n : n \in \mathbb{N} \rangle$.

- ▶ If we apply COH to $\mathcal{U}$ (which is, after all, a sequence of sets) we get an infinite set X such that for every n , either $X \subseteq^* U_n$ or $X \subseteq^* \overline{U_n}$.
- ▶ Then in X , as a subspace, we have that every basic open set is finite or cofinite.

Definition. A CSC space $\langle \mathbb{N}, \mathcal{U}, k \rangle$ is **stable** if every $U \in \mathcal{U}$ is finite or cofinite.

SGST₁: GST₁ restricted to stable spaces.

(Every stable T_1 CSC space has an infinite subspace that is discrete or has the cofinite topology.)

The above observation shows that over RCA_0 , $\text{SGST}_1 + \text{COH} \rightarrow \text{GST}_1$.

A decomposition theorem for GST_1

Theorem. Over RCA_0 , $\text{GST}_1 \leftrightarrow \text{SGST}_1 + \text{COH}$.

We show GST_1 implies ADS and hence COH. Let $\leq_L$ be a linear order on $\mathbb{N}$.

For all x , define $V_{x,x} = \{w : w \leq_L x\}$ and for $x \neq y$, define $V_{x,y} = V_{x,x} \setminus \{y\}$.

Let $\langle \mathbb{N}, \mathcal{U}, k \rangle$ be the CSC space generated by these sets. This is a T_1 space.

Apply GST_1 to get an infinite subspace Y .

- ▶ If Y has the cofinite topology then for every $x \in Y$ we have $V_{x,x} \cap Y =^* Y$, so x is $\leq_L$ -large in Y . Thus Y is an $\leq_L$ -descending sequence.
- ▶ If Y is discrete then for every $x \in Y$ there is $U \in \mathcal{U}$ such that $U \cap Y = \{x\}$. There is a finite set F such that $\{x\} = Y \cap \bigcap_{y \in F} V_{x,y}$. (Why?)
Thus $\{y \in Y : y \leq_L x\} \subseteq F$. Hence x is $\leq_L$ -small, so Y is $\leq_L$ -ascending.

SGST₁ and Σ_2^0 -Subset

D_2^2 : Every Δ_2^0 -definable $c : \mathbb{N} \rightarrow 2$ has an infinite homogeneous set.

Definition. Let Γ be a class of formulas. Γ -Subset contains, for each formula $\varphi(x) \in \Gamma$, the axiom $(\exists Y \text{ infinite})[(\forall x \in Y)\varphi(x) \vee (\forall x \in Y)\neg\varphi(x)]$.

$D_2^2 = \Delta_2^0$ -Subset. So in particular, Σ_2^0 -Subset $\rightarrow D_2^2$.

Theorem. Over RCA_0 , $\text{SGST}_1 \leftrightarrow \Sigma_2^0$ -Subset.

First, assume Σ_2^0 -Subset and let $\langle \mathbb{N}, \mathcal{U}, k \rangle$ be a stable T_1 space.

Let $\varphi(x) \equiv (\exists n)[U_n = \{x\}]$. By stability, $\neg\varphi(x) \rightarrow (\forall n)[x \in U_n \rightarrow U_n =^* \mathbb{N}]$.

Any infinite set Y such that $(\forall x \in Y)\varphi(x)$ forms a discrete subspace. Any infinite set Y such that $(\forall x \in Y)\neg\varphi(x)$ has the cofinite topology.

SGST₁ and Σ_2^0 -Subset

Conversely, let $\varphi(x)$ be a Σ_2^0 formula. Say $\varphi(x) \equiv (\exists y)(\forall z)\theta(x, y, z)$, $\theta \in \Sigma_0^0$.

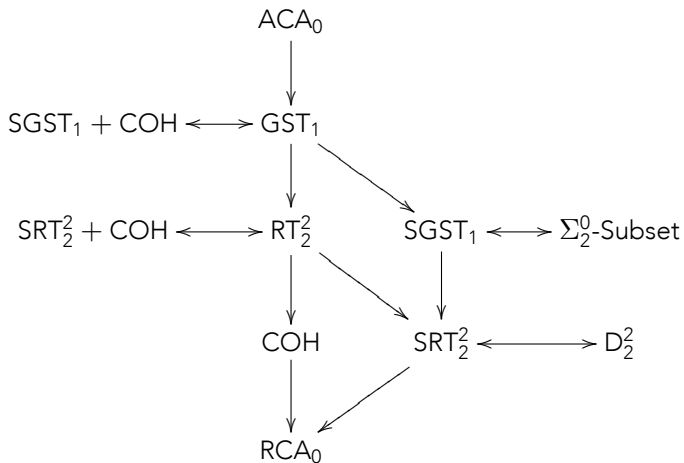
Define $V_{x,y,s} = \{x\} \cup \{w > \max\{x, s\} : (\exists z \leq w) \neg \theta(x, y, z)\}$.

- ▶ If $(\forall z)\theta(x, y, z)$ holds then $V_{x,y,s} = \{x\}$ for all s .
- ▶ Otherwise, there is a least z such that $\neg \theta(x, y, z)$ (Π_1^0)
and $V_{x,y,s} = \{x\} \cup \{w > \max\{x, s\} : w \geq z\}$ for all s .
- ▶ We have $\varphi(x)$ holds $\leftrightarrow V_{x,y,s} = \{x\}$ for some y and some (or all) s .

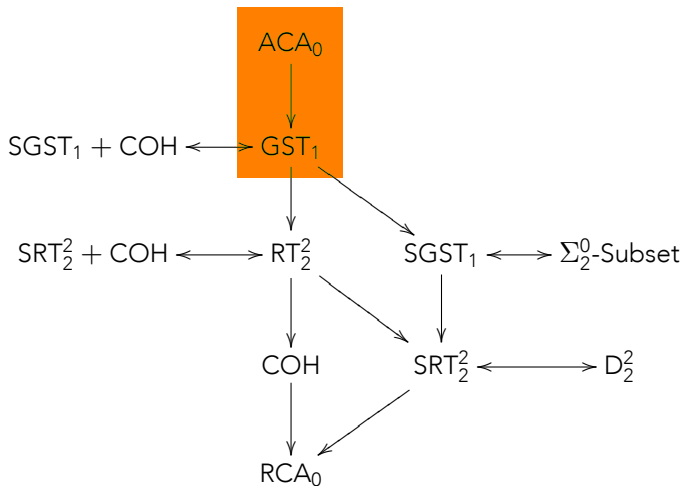
Let $\langle \mathbb{N}, \mathcal{U}, k \rangle$ be the CSC space generated by these sets. It is stable and T_1 .

- ▶ $\varphi(x) \rightarrow \{x\} \in \mathcal{U}$.
- ▶ $\neg \varphi(x) \rightarrow$ each $U \in \mathcal{U}$ with $x \in U$ is intersection of **uniformly cofinite** sets.
(In general, proving finite intersection of cofinite sets is cofinite requires $B\Sigma_2^0$.)

Where we are



Where we are



Cone avoidance and strong cone avoidance

A problem P admits **cone avoidance** if, for every $C \not\leq_T \emptyset$ and every $B \not\leq_T C$, every P -instance $X \leq_T B$ has a solution Y so that $B \oplus Y \not\leq_T C$.

Theorem (CJS). COH admits cone avoidance.

A problem P admits **strong cone avoidance** if, for every $C \not\leq_T \emptyset$ and every $B \not\leq_T C$, every P -instance X has a solution Y so that $B \oplus Y \not\leq_T C$.

(No effectivity restrictions on X , w.r.t. C or otherwise.)

Theorem (Dzhafarov & Jockusch). RT_2^1 admits strong cone avoidance.

Corollary. D_2^2 and Σ_2^0 -Subset admit strong cone avoidance.

An instance of D_2^2 or Σ_2^0 -Subset is just a formula $\varphi(x)$. Define $d : \omega \rightarrow 2$ by $d(x) = 1 \leftrightarrow \varphi(x)$. An infinite homogeneous set for d is a solution to φ .

Cone avoidance and strong cone avoidance

$D_2^2 \leftrightarrow \text{SRT}_2^2$. D_2^2 admits strong cone avoidance.

SRT_2^2 does **not** admit strong cone avoidance.

- ▶ Define $c : [\omega]^2 \rightarrow 2$ by $c(x, s) = 1 \leftrightarrow \emptyset'[s] \upharpoonright x = \emptyset' \upharpoonright x$.
- ▶ $\lim_y c(x, y) = 1$ for all x . (So ω is limit homogeneous for c .) If H is an infinite homogeneous set for c then $\emptyset' \upharpoonright x = \emptyset'[y] \upharpoonright x$ for all $x < y$ in H .

SRT_2^2 admits cone avoidance.

- ▶ Fix a $C \not\leq_T \emptyset$, $B \not\leq_T C$, and an B -computable stable $c : [\omega]^2 \rightarrow 2$. Define $d : \omega \rightarrow 2$ by $d(x) = \lim_y c(x, y)$. So $d \leq_T B'$.
- ▶ Let L be an infinite homogeneous set for d such that $B \oplus L \not\leq_T C$. By thinning, obtain $H \subseteq L$ homogeneous for c .
- ▶ We have $H \leq_T c \oplus L \leq_T B \oplus L$, so $B \oplus H \not\leq_T C$.

A metatheorem for non-implications

Theorem (folklore).

If $P_0, \dots, P_k$ admit cone avoidance then there exists an ω -model of $\text{RCA}_0 + P_0 + \dots + P_k + \neg \text{ACA}_0$.
(Iterate and dovetail argument.)

We build $\emptyset = B_0 \leq_T B_1 \leq_T \dots$ and let $\mathcal{M} = \{Y : (\exists n)[Y \leq_T B_n]\}$.
(This is a Turing ideal, so $\mathcal{M} \models \text{RCA}_0$.)

Suppose we have define $B_n \not\leq_T \emptyset'$ for some n . Say $n = \langle e, m, j \rangle$. ($m < n$.)

- Ask: is $j \leq k$ and is $\Phi_e^{B_m}$ a P_j -instance? If not, set $B_{n+1} = B_n$.
- If so, let Y be a solution $\Phi_e^{B_m}$ such that $B_n \oplus Y \not\leq_T C$. Set $B_{n+1} = B_n \oplus Y$.

Now if $X \in \mathcal{M}$ is any P_j -instance then $X = \Phi_e^{B_m}$ for some m . There is a solution $Y \leq_T B_{\langle e, m, j \rangle}$, hence $Y \in \mathcal{M}$. So $\mathcal{M} \models P_j$.

Finally, $\emptyset' \not\leq_T B_n$ for all n , hence $\emptyset' \notin \mathcal{M}$. So $\mathcal{M} \not\models \text{ACA}_0$.

Application to RT_2^2 and COH

COH, D_2^2 , and Σ_2^0 -Subset admit cone avoidance.

Theorem (Seetapun). Over RCA_0 , $RT_2^2 \not\vdash ACA_0$, even over ω -models.

$$(RT_2^2 \leftrightarrow D_2^2 + COH.)$$

Theorem Over RCA_0 , $GST_1 \not\vdash ACA_0$, even over ω -models.

$$(GST_1 \leftrightarrow \Sigma_2^0\text{-Subset} + COH.)$$



Mathias forcing

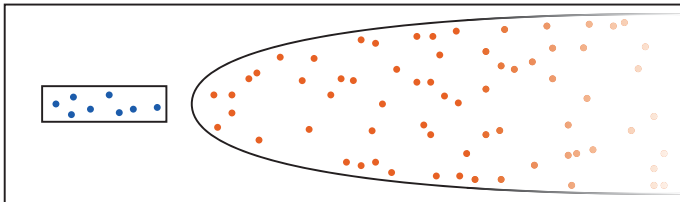
A **Mathias condition** is a pair (D, I) such that:

- ▶ D is a finite set;
- ▶ I is an infinite set with $\max D < \min I$.
(I is called a **reservoir**.)

A condition (E, J) **extends** (D, I) (written $(E, J) \leq (D, I)$) if:

- ▶ $D \subseteq E \subseteq D \cup I$;
- ▶ $J \subseteq I$.

A set G **satisfies** (D, I) if $D \subseteq G \subseteq D \cup I$.



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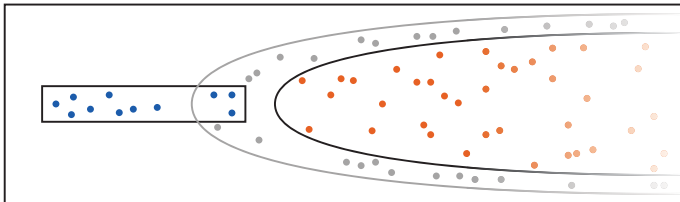
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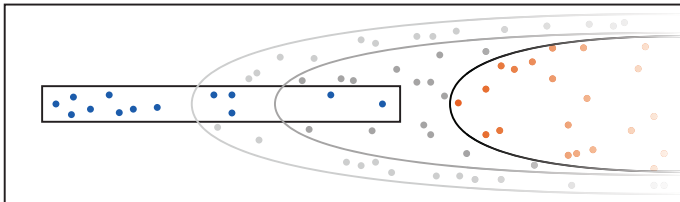
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An example: cohesive sets

Given any collection $\{\mathcal{C}_n : n \in \omega\}$ of **dense sets of conditions**, there exist $(D_0, I_0) \geq (D_1, I_1) \geq \dots$ so that $(\forall n)(\exists m)[(D_m, I_m) \in \mathcal{C}_n]$.

(This is a **generic sequence** for $\{\mathcal{C}_n : n \in \omega\}$. $G = \bigcup_m D_m$ is a **generic set**.)

Examples:

- ▶ For every n , $\mathcal{V}_n = \{(D, I) : |D| > n\}$ is dense. Indeed, for every condition (D, I) , if we let F consist of the least n elements of I then $(D \cup F, I \setminus F) \in \mathcal{V}_n$.
- ▶ If $\vec{R} = \langle R_n : n \in \omega \rangle$ is a sequence of sets then for every n , the set $\mathcal{W}_n = \{(D, I) : I \subseteq R_n \vee I \subseteq \overline{R_n}\}$ is dense.
- ▶ Indeed, for every condition (D, I) , either $I \cap R_n$ or $I \cap \overline{R_n}$ is infinite. So either $(D, I \cap R_n)$ or $(D, I \cap \overline{R_n})$ belongs to $\mathcal{W}_n$.

So every **sufficiently generic** set G is infinite and $\vec{R}$ -cohesive (for a given $\vec{R}$).

Cone avoidance of COH

Theorem (CJS). COH admits cone avoidance.

Fix $C \not\leq_T \emptyset$, a $B \not\leq_T C$, and an B -computable sequence $\vec{R} = \langle R_n : n \in \omega \rangle$.
For ease of notation, we assume $B = \emptyset$.

We build a set G that meets the following requirements for all n and e :

- ▶ $\mathcal{Q}_n : |G| \geq n$;
- ▶ $\mathcal{R}_n : G \subseteq^* R_n \vee G \subseteq^* \overline{R_n}$;
- ▶ $\mathcal{S}_e : \Phi_e^G \neq C$. (More generally, $\Phi_e^{B \oplus G} \neq C$.)

We use Mathias forcing with **computable** reservoirs.

(Conditions are (D, I) with $I \leq_T \emptyset$. More generally, $I \leq_T B$.)

We take G to be sufficiently generic for this notion of forcing.

Cone avoidance of COH

Let e and (D, I) be arbitrary. Ask: are there finite $F_0, F_1 \subseteq I$ and an x such that

$$\Phi_e^{D \cup F_0}(x) \downarrow \neq \Phi_e^{D \cup F_1}(x) \downarrow. \quad (\text{e-split over } D)$$

- ▶ If so: one of the two computations differs from $C(x)$, say $\Phi_e^{F_0}(x)$. Then $(D \cup F_0, I \setminus \varphi_e^{D \cup F_0}(x))$ is an extension of (D, I) forcing that G satisfies $\mathcal{S}_e$.
(Here $\varphi_e^{D \cup F_0}(x)$ is the **use** of $\Phi_e^{D \cup F_0}(x)$.)
- ▶ If not, we claim for no set S satisfying (D, I) is Φ_e^S total and equal to C .
(And so in particular, this is the case for our G , meaning $\mathcal{S}_e$ is satisfied.)
- ▶ Suppose otherwise. Then for every x we can **uniformly** find a finite $F \subseteq I$ such that $\Phi_e^{D \cup F}(x) \downarrow$. By assumption, we have $\Phi_e^{D \cup F}(x) = \Phi_e^S(x) = C(x)$.
- ▶ But then $C \leq_T I \leq_T \emptyset$, contradiction.

Strong cone avoidance of RT_2^1

Theorem (Dzhafarov & Jockusch). RT_2^1 admits strong cone avoidance.

We again fix $C \not\leq_T \emptyset$ and $B \not\leq_T C$. For ease of notation, take $B = \emptyset$.

Let $d : \omega \rightarrow 2$ be arbitrary and write $A_i = \{x : c(x) = i\}$.

Seeking a contradiction, suppose neither A_i has an infinite subset $G \not\leq_T C$.

We build **two sets** G_0 and G_1 that meet the following for all n, e_0, e_1 :

- ▶ $\mathcal{Q}_n : |G_0| \geq n \wedge |G_1| \geq n$;
- ▶ $\mathcal{R} : G_0 \subseteq A_0 \wedge G_1 \subseteq A_1$;
- ▶ $\mathcal{S}_{e_0, e_1} : \Phi_{e_0}^{G_0} \neq C \vee \Phi_{e_1}^{G_1} \neq C$.

(NB this is a disjunction!)

$\mathcal{S}_{e_0, e_1}$ implies $(\forall e)[\Phi_e^{G_0} \neq C]$ or $(\forall e)[\Phi_e^{G_1} \neq C]$. So one of G_0 or G_1 works.

(Why?)